PARKTOWN BOYS' HIGH SCHOOL

Department of Physical Sciences

PRELIMINARY EXAM
September 2023

Subject	Physical Sciences	**Marks**	150
Grade	12	**Topic**	Chemistry Memo

Question 1: Multiple Choice **(20)**

1.1 D ✓✓

1.2 C ✓✓

1.3 B ✓✓

1.4 D ✓✓

1.5 C ✓✓

1.6 A ✓✓

1.7 D ✓✓

1.8 A ✓✓

1.9 D ✓✓

1.10 D ✓✓

Q1 20 Own./Lisa.
Q2 23
Q3 13
Q4 14 Irma
Q5 14
Q6 18
Q7 21 Lisa.
Q8 17
Q9 11

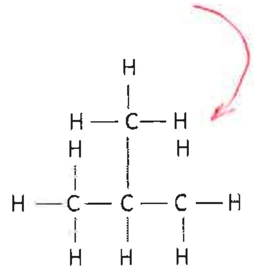
Question 2: Organic Molecules (23)

2.1.1 D ✓ (1)

2.1.2 C ✓ (1)

2.2 Propyne ✓ (1)

2.3 ~~2-methyl propane~~. (2)



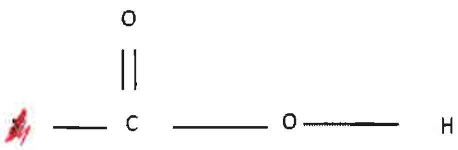
✓✓

2.4 Balanced chemical equation : 2 CH_4 + 13O_2 → 8 CO_2 + 10H_2O ✓✓✓✓ (4)

$2\ C_4H_{10} + 13\ O_2 \longrightarrow 8\ CO_2 + 10\ H_2O$

2.5.1 Esters ✓ (1)

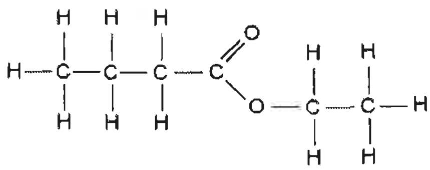
2.5.2 (2)



✓ 1

2.5.3 Butanoic Acid ✓✓ (2) 1

2.5.4 (2)



✓✓ (2)

2.6.1	H	C	O	(6)
	9,81	58,5	31,3	
$n = \frac{m}{M} = \frac{9,81}{1}$		$= \frac{58,85}{12}$	$= \frac{31,3}{16}$	
	9,81✓	4,9 ✓	1,956 ✓	
	$\frac{9,81}{1,956}$	$\frac{4,9}{1,956}$	$\frac{1,956}{1,956}$	
	5 :	2.5 :	1	
	10 :	5	2✓	

Molar mass : 10(1) + 5(12) + 2(16) = 102g.mol

= 102/102

= 1✓

∴ $C_5H_{10}O_2$✓

(6)

2.5.2 propanoic acid was the other compound. ✓ (1)

Question 3 Intermolecular Forces : (13)

3.1.1 Carboxylic Acids ✓ (1)
3.1.2 Alcohols ✓ (1)
3.2.1 Aldehydes ✓ (1)
3.2 Functional Group : A bond/atom or group of atoms that determines the physical and chemical properties of a group of organic compounds. ✓✓ (2)

3.3.1 Propanal ✓ (1)
3.3.2 Pentan-1-ol ~~or pentanol~~ ✓ (1)
3.4 ~~Homologous Series/Functional Group~~ ✓ Mass (1)

3.5 Boiling point of carboxylic acids ⟩ alcohols ⟩ aldehydes ✓✓ (2)

3.6 ✓

- Size of the molecule increases with additional carbon group. ✓
- Mass increases as does van der Waals forces ✓
- More energy is required to overcome the intermolecular forces. ✓

(3)

Fg

Question 4 : Organic Flow Charts (14)

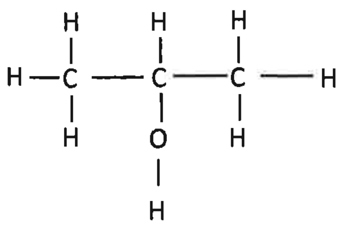
4.1. Contains a double bond between C - atoms (1)

4.2

(3)

4.3 Addition or Hydrobromination ✓ | Hydrohalogenation (1)

4.4 Propan-2-ol (3)



3

4.5 Hydrolysis✓ (1)

4.6.1 Water ✓ (1)

4.6.2 $H_3PO_4/H_2SO_4/HC\ell/HBr$ ✓ (1)

4.6.3 Addition/ Hydration (1)

4.7.1 Propene (1)

4.7.2 Elimination/ Dehydrohalogenation (1)

Question 5 Rates of Reactions **(14)**

5.1 Smaller Than ✓ (1)

5.2 To act as a control and to make the test fair ✓ (1)

5.3 The same mass/volume of Mg will react with the same amount of acid ✓ Mg is a limiting reagents. ✓ (2)

5.4.1 60 cm^3 ✓ (1)

5.4.2 42 cm^3 ✓ ✓ (2)

5.5 Experiment 1 ✓ : gradient of the graph is steeper. ✓ (2)

5.6 Reaction rate increases with increase in concentration ✓✓ (2)

NOTE : -1 if per unit time is mention in the same sentence if rate is used

5.7.1 Remains the same ✓ (2)

5.7.2 Increases ✓ (2)

Question 6 : Equilibrium : **(18)**

6.1 When the equilibrium in a closed system is disturbed ✓ the system will shift the equilibrium position **OR** re-instate a new equilibrium as to **OR** favour the reaction that will oppose **OR** cancel **OR** counteract the change **OR** disturbance. ✓

OR / *OF*

When a stress / change is placed on a system in equilibrium ✓
The system shifts the equilibrium (position) **OR** re-instate a new equilibrium √ so as to remove **OR** cancel **OR** oppose the stress / change. ✓

OR / OF

When the conditions affecting an equilibrium are changed, ✓
the equilibrium (position) shifts in such a way as to oppose the change **OR** cancel the change. ✓ (2)

6.2 At t =120s volume/pressure increases. ∴ reactants and products increase in concentration✓ Reverse reaction is favoured✓ to form less mol gas✓ (3)

decreases

6.3 Exothermic : ✓
Stress : decrease in temperature, reverse reaction decreases. Forward reaction is favoured to raise the temperature✓ (2)

6.4.1 No effect ✓ (1)

6.4.2 Increases – must be consistent with question 6.3✓ (1)

6.5.1 $V = \frac{n}{c}$

$= \frac{3,6}{1,2}$✓

$= 3\ dm^{-3}$✓ (2)

6.6.2 ***CALCULATIONS USING NUMBER OF MOLES***

Mol Table	**$2A_2B$**	**$2A_2$**	**B_2**
Mol ratio	2	2	1
Mol at start	5,1	0	0
Mol used	3,6	3,6	1,8
Mol at equilibrium	1,5 (5,1 - 3,6) √	3,6	1,8
Concentration at equilibrium	0,5 √	1,2	0,6

OR

CALCULATIONS USING CONCENTRATION

Mol Table	$2A_2B$	$2A_2$	B_2
Mol ratio	2	2	1
Concentration at start	1,7 (5,1/3) √	0	0
Concentration change	1,2	1,2	0,6
Concentration at equilibrium	0,5 (1,7 – 1,2) √	1,2	0,6

Or using formulae :

$[A_2B] = \frac{n}{v}$

$[A_2B] = \frac{5,1-3,6}{3}$

$= 0,5\ mol.dm^{-3}$

$[B_2] = \frac{n}{v}$

$[B_2] = \frac{1,8}{3}$

$= 0,6\ mol.dm^{-3}$

$$K_C = \frac{[A_2]^2[B_2]}{[A_2B]^2}$$

$$= \frac{1,2^2 \times 0,6}{0,5^2}\ ✓$$

$K_C = 3,46$ ✓ -1 IF UNITS ARE PRESENT (6)

Question 7 Acids and Bases ***(21)***

7.1.1 The reaction of a molecular substance with water to produce ions ✓✓ (2)

7.1.2 A proton acceptor ✓ (1)

7.1.3 Sulphurous acid ✓ (1)

7.1.4 It partially ionises is solution. ✓ HSO_3^- is a weak acid. ✓ (2)

7.2 The $[SO_3^{2-}]$ will decrease ✓
The reaction that favours the SO_3^{2-} production will be initiated✓
The forward reaction is faoured✓
$[H3O^+]$ increases ∴ pH decreases✓ (5)

7.3.1 $n\ (H_2SO_4) = CV$✓

$= 1 \times 0,05$✓

$= 0,05\ mol$ ✓ (3)

7.3.2 $n\ (NaOH) = CV$

$= 0,5 \times 0,028$✓

= 0,014 mol ✓

RATIO : 2 : 1

0,007 : 0,014

Coe from 7.3.1 0,05 – 0007 = 0,043✓

RATIO 1:1✓

m ($MgCO_3$) = nM

0,043 x 84,3✓ = 3,62g✓

% purity = 3,62/5 x 100 = 72,5%✓ (7)

Question 8 Galvanic Cellls *(17)*

8.1.1 A : $C\ell_2$✓

B : $C\ell^-$ ✓ (2)

8.1.2 C : Pt✓

D : Fe^{2+}; Fe^{3+} ✓ (2)

8.2.1 Salt Bridge completes the circuit ✓ or maintains electrical neutrality✓ (1)

8.2.2 Any group 1 metal (KNO_3) ✓ (1)

8.2.3 Ions present allow electrical conductivity or highly soluble and inert for spectator ion. (2)

8.2.4 Exothermic - spontaneous reaction that releases energy✓✓ (2)

8.3 E^θ cell = E^θ cathode – E^θ anode ✓

= 1,36✓ – (0,77)✓

= 0,59 V ✓ (3)

8.4 Fe^{3+} will decrease more than Fe^{2+} ✓

Forward reaction will be favoured ✓ to increase the concentration of the Fe^{3+} ✓

EMF will increase ✓ (4)

Question 9 Electrolytic cell *(11)*

9.1 The process in which electricity is used to bring about a chemical change/ decompose/ break compounds into components ✓✓

OR

A process where electrical energy ✓is converted into chemical energy. ✓ (2)

9.2 P ✓

- P is the anode✓
- Oxidation takes place at the anode✓ (3)

9.3 $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$ ✓✓ (2)

9.4 Pt and Ag are both weaker reducing agents and will not undergo oxidation ✓✓

Or

Cu is a stronger reducing agent and will be oxidised. ✓✓ (2)

9.5 The rate at which copper is oxidised at the anode is equal to the rate at which copper ios are reduced